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Uncertainty Quantification of 3D Stochastic Fault Modeling in Structurally Complex Reservoirs

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Abstract

Faults play a pivotal role in the characterization of oil and gas reservoirs. Especially in heavily faulted reservoirs, faults have significant effects on the estimation of reservoir volumes since they define the boundaries of the reservoir and compartmentalize the reservoirs into distinct fault-bounded blocks. This paper presents a method utilizing stochastic simulation of the 3D faults modeling to quantify the uncertainties and minimize their impacts.

Fault interpretation uncertainties are represented by creating fault uncertainty envelope volumes that encompass both the footwall and hanging wall sides of the "most likely" base-case fault surfaces. At all interpreted fault points, lateral and vertical uncertainty are defined based on the quality of seismic data and interpretation confidence level. 3D fault uncertainties modeling involves generating hundreds of 3D stochastic fault models to quantify uncertainty. Key fault geometrical parameters that are perturbed include fault lateral position, fault dip, fault strike and fault throw. Inside the fault uncertainty envelope volumes, these fault uncertainty parameters are stochastically sampled using a triangular probability distribution.

3D uncertainties modeling extends beyond faults to horizons, fluid contacts, and the entire model, ensuring preservation of the 3D geological model's integrity. The geometry of the 3D geological grid is updated concurrently with the re-distribution of reservoir properties onto the 3D grid. Moving faults and horizons obviously alters the reservoir volumes. Consequently, bulk and hydrocarbon volumes above fluids contacts are recalculated for each simulation run. Through the uncertainty runs, the asset team was able to measure the effect and sensitivity of individual fault parameter interpretations on reservoir volumes. By ranking hundreds of structural uncertainty scenarios based on reservoir volume, the corresponding 3D geomodels linked to P10, P50, and P90 values were identified and prepared for submission to dynamic flow simulation studies.

This approach moves beyond conventional methods that assume simple scalar uncertainties away from wells control and deterministic faults, providing a comprehensive representation of 3D structural framework uncertainties. It enables the quantification of the impact and sensitivity of faults parameters on reservoir volumes within heavily fault-bounded reservoirs.

Introduction

The accurate representation of faults in 3D geological models is critical for hydrocarbon exploration and development, especially in structurally complex reservoirs. Faults affect both static and dynamic reservoir characteristics. Structurally, they define the geometry of the subsurface, large effects on the estimation of reservoir volumes, and compartmentalize reservoirs into separate fault blocks. Dynamically, they act as potential flow barriers or conduits, impacting pressure communication and hydrocarbon movement.

In seismic interpretation workflows, faults are often treated deterministically based on the interpreter's best guess guided by seismic reflections. However, multiple interpretations of the same data can be equally probable, particularly where seismic quality is poor or complex faulting obscures reflector continuity. However, conventional 3D structural modeling workflows inadequately account for uncertainty in fault location and geometry. This can lead to misrepresentation of fault-bounded compartments, inaccurate volume estimation, and flawed well placement strategies.

The need for a probabilistic approach in 3D structural and fault modeling has become more important as fields mature and marginal developments demand tighter risk management. Integrated 3D fault uncertainty quantification methods are gaining more attention, but they have traditionally focused on horizon modeling and reservoir property distributions. Fault uncertainty remains a relatively underdeveloped subject, despite its impact on volumetrics and compartmentation.

Fault modeling uncertainty arises from multiple sources (Thore, et al., 2002). These include seismic acquisition issues (e.g., low signal-to-noise ratio), processing artifacts (e.g., migration errors), and interpretation ambiguities (e.g., mispicking fault terminations or misaligning reflections). Time-to-depth conversion introduces additional uncertainty, especially when velocity models are poorly constrained. Even subtle variations in dip or lateral fault placement can have substantial volumetric consequences when faults cut across the most prolific hydrocarbon zones and act a bounding surface.

To address these limitations, this paper presents a robust fault stochastic workflow that explicitly represents and propagates fault parameter uncertainties through the integrated 3D structural model. By doing so, it enables geoscientists and geomodeler to better understand the range of possible subsurface scenarios and their impact on reservoir modeling.

This paper focuses on a case study involving a highly faulted reservoir with complex fault geometries and compartmentalized structure. We demonstrate the application of stochastic fault uncertainty modeling, its integration with horizon and fluid contact uncertainty, and the resulting understandings into possible volumetric variability and sensitivity. The resulting 3D structural framework models provide a realistic foundation for reservoir simulation and development planning, enabling more informed, data-driven decisions in the presence of fault modeling uncertainty.

Method

The stochastic fault modeling workflow presented in this study begins by defining the uncertainty associated with seismic fault interpretations. Given that seismic data inherently brings limitations due to resolution, acquisition geometry, and noise, the first step involves defining the spatial bounds of uncertainty for each interpreted point of faults and horizons. These bounds are established by constructing 'fault uncertainty envelopes'—volumetric zones within which the most likely base case fault position is allowed to reside (Yang et al., 2013, Røe et al., 2014).

These envelopes are built by assessing both lateral and vertical uncertainties at each interpreted point. The magnitude of uncertainty is influenced by the quality of seismic data, confidence in interpretation, structural complexity, and proximity to well control. A triangular probability distribution is used to represent the relative likelihood of various fault positions, giving higher probability to those close to the interpreted surface and linearly decreasing toward the edges of the envelope.

Fault uncertainty envelopes are modeled for all major and minor faults in the 3D structure fault model. Fault geometrical parameters, including lateral position, dip, strike, and throw are perturbed independently within the defined uncertainty ranges. These perturbed parameters are then used to stochastically generate hundreds of fault realizations. Each realization represents a possible fault configuration, following structural constraints such as fault truncation rules and structural hierarchy.

Simultaneously, uncertainties in horizon interpretations and fluid contacts are incorporated to ensure consistency and the integrity across the entire 3D model. For each realization, surfaces are adjusted accordingly, and the 3D geological grid is regenerated. This dynamic updating process ensures that the geometric and petrophysical integrity of the model is preserved throughout the 3D geomodel.

Once 3D structural models are built and 3D reservoir properties are updated, volumetric calculations are conducted for each realization. Bulk rock volume, gross rock volume, net-to-gross ratio, facies, porosity, water saturation, and hydrocarbon pore volume are regenerated and recalculated for each stochastic model. These calculations provide understanding into how structural uncertainty propagates through to an integrated 3D reservoir uncertainties modeling workflow.

Sensitivity analysis is applied using statistical tools such as Tornado charts to identify the most influential parameters on volumetric estimation. This process helps isolate specific fault parameters—such as throw variation on a particular fault block—that have a substantial impact on reservoir volume. This insight is critical for risk mitigation and decision-making in development planning.

The method is implemented using a commercial geological modeling software, combining scripting and workflow management for automation, and statistical tools for uncertainty sampling. The fault models are constructed in a structural modeling module, while the uncertainty distribution and stochastic sampling are implemented using a geostatistical algorithm module. Post-processing, including 3D reservoir model realization retrieval and visualization, is performed using this commercial 3D modeling software.

This comprehensive methodology ensures that every significant source of fault and horizon structural uncertainty is considered and quantitatively propagated through the full integrated 3D stratigraphic and structural framework geomodel. The result is a probabilistic framework that supports the generation of realistic, structurally valid scenarios ready for reservoir modeling and optimization workflows.

Results and Discussion

The application of the 3D stochastic fault uncertainty modeling workflow to a structurally complex reservoir has produced insightful results regarding the spatial variability of fault interpretations and their impact on reservoir volumetrics. A total of 250 structurally consistent 3D fault models were generated within predefined uncertainty envelopes. Each realization was assessed for consistency in fault geometry, truncation rules, and preservation of the overall 3D structural framework and 3D reservoir properties.

Figure 1 illustrates a representative structural cross-section where five key faults and two key seismic reference horizons have been simulated across 250 stochastic realizations. All fault surfaces lie within the bounds of their respective uncertainty envelopes, demonstrating the effectiveness of the envelope constraints in bounding possible geological variability. The internal layering between key horizons was constructed using proportional layering, and well control was exactly honored.

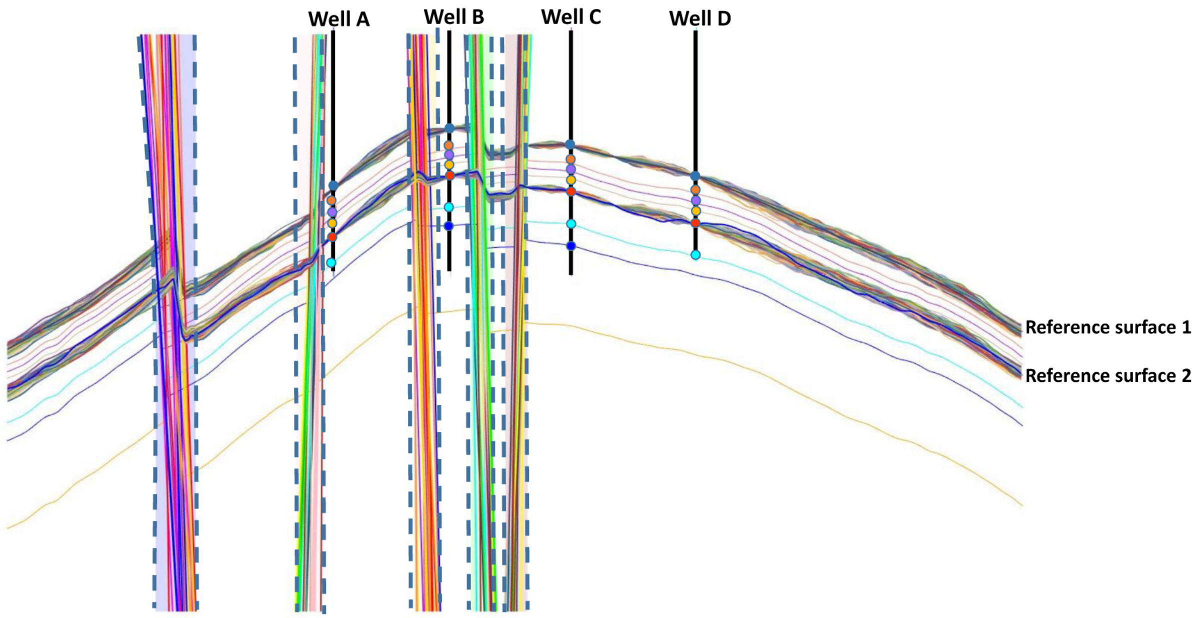

Figure 1—A cross section showing 250 stochastic realizations of five fault surfaces and two key reference surfaces. Internal surfaces were proportionally generated using both reference surfaces and well tops. The dotted lines represent the edges of the fault envelope. All realizations of fault surfaces were bounded inside the fault envelope.

Map views of fault line realizations (Figure 2) show a wide lateral distribution of fault traces, especially in structurally complex zones where seismic quality is lower. The spread of fault positions reflects the interpreter's confidence levels and seismic data quality input into the uncertainty definition. In some areas, the fault position deviates by more than 100 meters from the base case, which could have significant implications for reservoir connectivity and well path design.

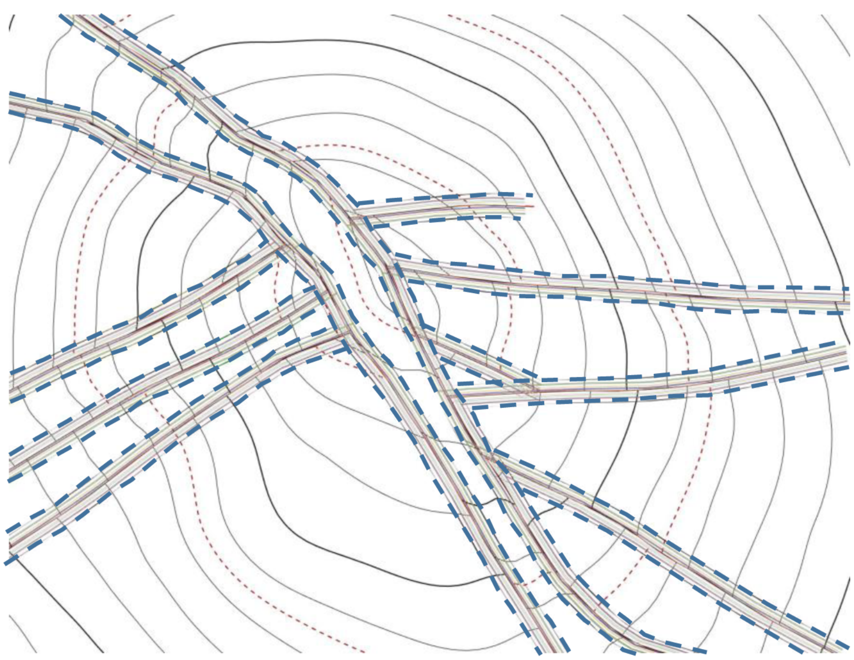

Figure 2—A map view of 250 fault line realizations bounded inside the fault uncertainty envelope.

In Figure 3, we present 3D geomodel realizations related to gross reservoir volume scenarios at P10, P50, and P90 levels. Each of these realizations shows lateral and vertical displacement of key faults, resulting in varied fluid contact regions and structural closures. The crestal region of the reservoir, which is particularly sensitive to fault position, shows the most variability in volumetric outcome. Vertical sections confirm how minor shifts in fault dip and strike influence the trap geometry and hydrocarbon and fluid distribution.

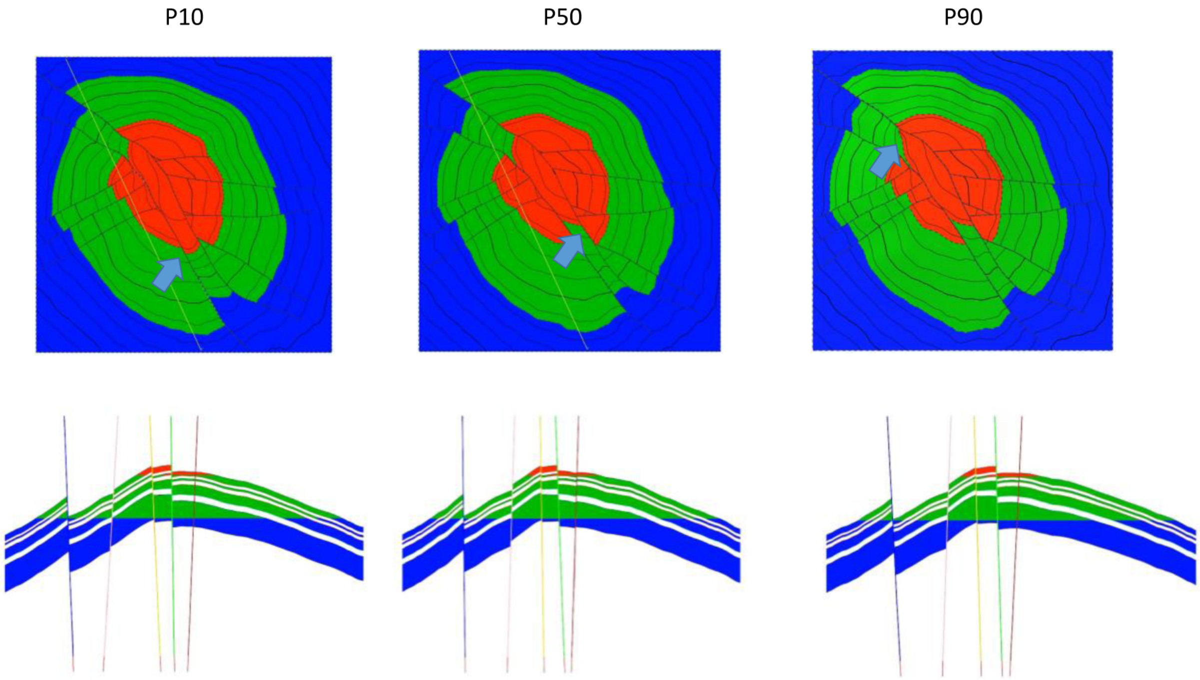

Figure 3—a) Map views of the first K-layer of 3D geologic grid overlaid with fluid contact regions of realization corresponding to P10, P50 and P90 gross reservoir volume. The arrow indicates the location where some fault positions are laterally moved in the crestal area. b) Vertical structural cross section showing fault and horizons realizations related to P10, P50 and P90 gross volume.

Volumetric recalculations for each realization included gross rock volume, net rock volume, net-to-gross ratio, and hydrocarbon pore volume above the fluid contacts. Histogram plots (Figure 4) of the bulk gas volume distribution reveal a near-normal distribution, with standard deviation quantifying the volumetric uncertainty driven by structural and fault variability. A cumulative probability plots further aids in extracting probabilistic volume estimates, identifying models corresponding to P10, P50, and P90 scenarios suitable for flow simulation.

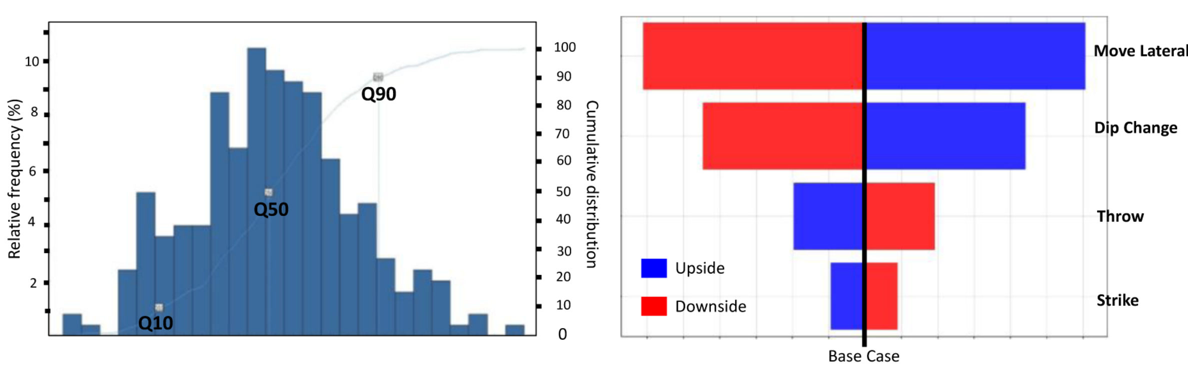

Figure 4—Histogram and cumulative distribution of bulk gas volume computed with more than 250 realizations of the structural model. Tornado chart shows the sensitivity of fault uncertainty parameters of one major fault to reservoir volume.

A Tornado sensitivity chart highlights the relative impact of each fault uncertainty parameters on volumetric uncertainty. The fault lateral shift contributed more to volumetric variability than fault dip, fault throw, or fault dip changes. This finding highlights the importance of targeted interpretation refinement in sensitive parameter rather than pursuing uniform enhancements across the fault interpretation and its model.

The integrated reservoir characterization and reservoir management team found these outputs critical in field development planning. For instance, wells previously planned near fault intersections were re-evaluated using high P10 and P90 displacement realizations to assess drilling risk and potential reservoir compartmentalization.

The results also provided insights into the correlation between interpretation uncertainty and seismic data quality. Areas with wider envelope widths coincided with zones of poor signal-to-noise ratio or complex fault shadows. This feedback helped guide future seismic acquisition and reprocessing efforts to reduce interpretation uncertainty.

In summary, the results validate the methodology's ability to represent 3D fault and structural uncertainty realistically and quantify its impact on key reservoir volumetrics. More importantly, they demonstrate the value of probabilistic 3D fault and structural modeling in bridging the gap between interpretation uncertainty and the reservoir model. The approach helps to reduce development risk, optimize well placement, and improve the overall robustness of reservoir modeling and forecasts.

Conclusion

This study provides a comprehensive approach to quantify uncertainty in fault interpretations through 3D stochastic fault and structural modeling. The workflow accounts for the inherent key uncertainties in fault geometry—such as position, dip, strike, and throw—using fault uncertainty envelopes and probabilistic sampling. These uncertainties are propagated into a full 3D geological model that also includes variations in horizons and fluid contacts. The integration ensures structural consistency and 3D reservoir properties.

The modeling of hundreds of stochastic fault realizations has demonstrated how small changes in fault interpretation can lead to significant differences in volumetric estimation. By computing and comparing key fault uncertainties parameters across all realizations, the asset team gains an understanding of how fault and structural uncertainty can impact development planning. Furthermore, by applying sensitivity analysis, the workflow allows the asset team to identify the most influential fault uncertainty parameters.

The ability to visualize and extract P10, P50, and P90 cases provides a significant connection between static modeling uncertainty and reservoir simulation. This enables more informed decision-making when selecting scenarios for dynamic simulation, assessing reserves, and planning field development. Significantly, it helps key decisions such as well placement and resource assessment.

Incorporating this uncertainty modeling into standard interpretation workflows represents a shift from deterministic to probabilistic 3D fault modeling and the overall 3D structural framework modeling. As such, the methodology significantly enhances the robustness of reservoir models in structurally complex settings and supports a better strategy for volumetric resource estimation and field development under uncertainty.

Novelty and Additive Value

This paper presents a practical and comprehensive method for quantifying 3D fault and structural framework modeling uncertainties, specifically focused on fault geometry in complex reservoirs. Unlike traditional workflows that assume deterministic faults and consider uncertainty only in reservoir properties or horizons, this approach provides an integrated solution to capture fault uncertainty in all critical components of the structural model.

The use of fault uncertainty envelopes, coupled with stochastic simulation of fault geometries, allows for the realistic representation of interpretational uncertainties and its volumetric consequences. The simultaneous perturbation of faults, horizons, and contacts ensures structural integrity and improves the realism of the resulting models. This full uncertainty modeling—from interpretation to volumetric assessment—provides a powerful decision-support tool for reservoir management.

Furthermore, this methodology enables the construction of uncertainty-based models suitable for reservoir characterization, modeling and simulation, reserves estimation, and field development planning. The capability to link interpretation uncertainty directly to reservoir modeling is an advancement in reservoir modeling under uncertainty.

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